

P1.

(a). $a_n = 4^n + 2 \cdot 3^n$

$$A(X) = \sum_{n=0}^{\infty} a_n X^n = \sum_{n=0}^{\infty} (4^n + 2 \cdot 3^n) X^n$$
$$= \sum_{n=0}^{\infty} (4X)^n + 2 \sum_{n=0}^{\infty} (3X)^n$$

$$\therefore \sum_{n=0}^{\infty} X^n = \frac{1}{1-X}.$$

$$\therefore A(X) = \frac{1}{1-4X} + \frac{2}{1-3X}.$$

(b). $b_n = \binom{n+2}{2} 2^n$

$$B(X) = \sum_{n=0}^{\infty} b_n X^n = \sum_{n=0}^{\infty} \binom{n+2}{2} 2^n X^n$$

$$\therefore \frac{1}{(1-X)^3} = \left(\sum_{n=0}^{\infty} X^n \right)^3, \text{ consider } [X^n].$$

$$n = a+b+c \quad (a, b, c \in \mathbb{N}) \text{ has } \binom{n+2}{2} \quad (a, b, c).$$

$$\text{So } \frac{1}{(1-X)^3} = \sum_{n=0}^{\infty} \binom{n+2}{2} X^n.$$

$$\text{So } B(X) = \sum_{n=0}^{\infty} \binom{n+2}{2} (2X)^n = \frac{1}{(1-2X)^3}.$$

P2.

$$(a) \text{ Apple: } 1 + x^2 + x^4 + \dots + x^{2n} + \dots = \frac{1}{1-x^2}.$$

$$\text{Orange: } 1 + x + x^2.$$

$$\text{Banana: } 1 + x^3 + x^6 + \dots + x^{3n} + \dots = \frac{1}{1-x^3}.$$

$$\text{Pear: } 1 + x.$$

so the generating function is

$$f(x) = \frac{1}{1-x^2} \cdot (1+x+x^2) \cdot \frac{1}{1-x^3} \cdot (1+x).$$

$$= \frac{(1+x)(1+x+x^2)}{(1-x)(1+x) \cdot (1-x)(1+x+x^2)} = \boxed{\frac{1}{(1-x)^2}}$$

$$= \left(\sum_{n=0}^{\infty} x^n \right)^2.$$

consider $[x^n]$. $n = a+b$. ($a, b \in \mathbb{N}$) has $n+1$ ways.

$$\text{so } f(x) = \left(\sum_{n=0}^{\infty} x^n \right)^2 = \left[\sum_{n=0}^{\infty} (n+1) x^n \right].$$

$$(b) \quad f(x) = \sum_{n=0}^{\infty} (n+1) x^n \Rightarrow \boxed{a_n = n+1,} \quad \forall n \in \mathbb{N}.$$

P3.

(a) r parts odd, $k-r$ parts even.

$$\begin{aligned} f(x) &= (x + x^3 + x^5 + \dots + x^{2n+1} + \dots)^r \cdot (1 + x^2 + x^4 + \dots + x^{2n} + \dots)^{k-r} \cdot \binom{k}{r} \\ &= \frac{x^r}{(1-x^2)^r} \cdot \frac{1}{(1-x^2)^{k-r}} \cdot \binom{k}{r} \\ &= \frac{x^r}{(1-x^2)^k} \binom{k}{r} \end{aligned}$$

(b). $\frac{1}{(1-x)^k} = \left(\sum_{n=0}^{\infty} x^n \right)^k$, consider $[x^n]$.

$n = a_1 + a_2 + \dots + a_k$, ($a_i \in \mathbb{N}$) has $\binom{n+k-1}{k-1}$ ways.

$$\text{so } \frac{1}{(1-x)^k} = \sum_{n=0}^{\infty} \binom{n+k-1}{k-1} x^n.$$

$$\text{so } f(x) = \frac{x^r}{(1-x^2)^k} \binom{k}{r} = x^r \binom{k}{r} \sum_{n=0}^{\infty} \binom{n+k-1}{k-1} x^{2n}$$

consider $[x^n]$, $r+2m=n \Rightarrow m = \frac{n-r}{2}$.

so $a_n = \binom{k}{r} \binom{\frac{n-r}{2} + k - 1}{k-1}$, when $\frac{n-r}{2} \in \mathbb{N}$ and $n \geq r$.

otherwise $a_n = 0$.

In conclusion,
$$a_n = \begin{cases} \binom{k}{r} \binom{\frac{n-r}{2} + k - 1}{k-1}, & \text{when } \frac{n-r}{2} \in \mathbb{N} \text{ and } n \geq r. \\ 0 & \text{otherwise.} \end{cases}$$

so $a_n = 0$ when $\frac{n-r}{2} \notin \mathbb{N}$ or $n < r$

P4. $\sum_{j=0}^n \binom{r+j-1}{r-1} \binom{s+n-j-1}{s-1} = \binom{r+s+n-1}{r+s-1}$.

(1) Counting weak compositions.

Consider $n = a_1 + a_2 + \dots + a_r + b_1 + b_2 + \dots + b_s$, ($a_i, b_i \in \mathbb{N}$).

RHS: there are $\binom{n+r+s-1}{r+s-1}$ ways to choose $(a_1, a_2, \dots, a_r, b_1, b_2, \dots, b_s)$ in total.

LHS: First choose $a_1, a_2, \dots, a_r$, consider $j = \sum_{i=1}^r a_i$.

j can be $0, 1, 2, \dots, n$. there are $\binom{r+j-1}{r-1}$ ways to choose $(a_1, a_2, \dots, a_r)$.

For a specific j , let $b_1 + b_2 + \dots + b_s = n - j$,

there are $\binom{s+n-j-1}{s-1}$ ways to choose $(b_1, b_2, \dots, b_s)$.

by multiplication principle,

there are $\sum_{j=0}^n \binom{r+j-1}{r-1} \cdot \binom{s+n-j-1}{s-1}$ ways to choose $(a_1, a_2, \dots, a_r, b_1, b_2, \dots, b_s)$ in total

so LHS = RHS.

(2). Generative function.

$$\frac{1}{(1-x)^r} = \left(\sum_{n=0}^{\infty} x^n \right)^r, \text{ consider } [x^n].$$

$n = a_1 + a_2 + \dots + a_r$, $a_i \in \mathbb{N}$. there are $\binom{n+r-1}{r-1}$ ways.

$$\text{so } \frac{1}{(1-x)^r} = \sum_{n=0}^{\infty} \binom{n+r-1}{r-1} x^n$$

$$\text{so } \sum_{n=0}^{\infty} \binom{r+n-1}{r-1} x^n \cdot \sum_{n=0}^{\infty} \binom{s+n-1}{s-1} x^n = \frac{1}{(1-x)^r} \cdot \frac{1}{(1-x)^s} = \frac{1}{(1-x)^{r+s}} = \sum_{n=0}^{\infty} \binom{r+s+n-1}{r+s-1} x^n.$$

$$\text{by convolution, LHS} = \sum_{n=0}^{\infty} \sum_{j=0}^n \binom{r+j-1}{r-1} \cdot \binom{s+n-j-1}{s-1} x^n.$$

$$\text{compare } [x^n], \sum_{j=0}^n \binom{r+j-1}{r-1} \cdot \binom{s+n-j-1}{s-1} = \binom{r+s+n-1}{r+s-1}.$$

P5.

$$(a). f(x) = (1 + x + x^2 + \dots + x^b)^k \quad u_i \in \{0, 1, 2, \dots, b\}.$$

$$= \left(\frac{1 - x^{b+1}}{1 - x} \right)^k$$

$$(b). (1 - x^{b+1})^k = \sum_{i=0}^k \binom{k}{i} (-1)^i x^{(b+1)i}$$

$$\frac{1}{(1-x)^k} = \left(\sum_{n=0}^{\infty} x^n \right)^k = \sum_{n=0}^{\infty} \binom{n+k-1}{k-1} x^n.$$

$$\text{so } f(x) = \sum_{i=0}^k \binom{k}{i} (-1)^i x^{(b+1)i} \cdot \sum_{m=0}^{\infty} \binom{n+k-1}{k-1} x^m$$

$$\text{consider } [x^n]. \quad n = (b+1)i + m. \Rightarrow m = n - (b+1)i.$$

$$\text{so } a_n = \sum_{\substack{i \leq k \\ (b+1)i \leq n}} \binom{k}{i} (-1)^i \binom{n - (b+1)i + k - 1}{k - 1}$$

$$= \sum_{i=0}^{\min\{k, \lfloor \frac{n}{b+1} \rfloor\}} \binom{k}{i} (-1)^i \binom{n - (b+1)i + k - 1}{k - 1}$$

P6.

$$(a). \begin{cases} 0 \leq C_k \leq r \\ C_k \leq k-1 \end{cases} \Rightarrow 0 \leq C_k \leq \min\{r, k-1\}$$

$$\text{Let } \text{Inv}(w) = \sum_{k=1}^n C_k. \quad \text{by multiplication principle}$$

generating function:

$$f(x) = \prod_{k=1}^n (1 + x + x^2 + \dots + x^{\min\{r, k-1\}})$$

$$= (1 + x + x^2 + \dots + x^r)^{n-r} \prod_{k=1}^r (1 + x + x^2 + \dots + x^{k-1})$$

(b). Total number of permutations

$$= \sum_{j \geq 0} a_j = \text{the sum of coefficient in } f(x) = f(1).$$

$$f(1) = (r+1)^{n-r} \cdot r!$$

$$(c) r=1, \quad f(x) = (1+x)^{n-1} \cdot 1 = (1+x)^{n-1}$$

$$\text{so } a_j = \binom{n-1}{j}, \quad 0 \leq j \leq n-1, \quad \text{and } a_j = 0, \quad j \geq n$$

$$\text{so } a_j = \begin{cases} \binom{n-1}{j}, & 0 \leq j \leq n-1 \\ 0, & j \geq n. \end{cases} \quad \begin{array}{l} a_j \text{ means the number of} \\ \text{permutations in } S_n \text{ having exactly} \\ j \text{ inversions.} \end{array}$$

P7.

$$(a) \omega = 35142 \Rightarrow \omega(1)=3, \omega(2)=5, \omega(3)=1, \omega(4)=4, \omega(5)=2.$$

$$\Rightarrow \omega^{-1}(1)=3, \omega^{-1}(2)=5, \omega^{-1}(3)=1, \omega^{-1}(4)=4, \omega^{-1}(5)=2.$$

$$\text{so } D(\omega) = \{(1,1), (1,2), (2,1), (2,2), (4,2)\}.$$

$\omega^{-1}(j)$	j	$\omega(i)$	3	5	1	4	2
3	1	1	✓	✓	×	×	×
5	2	2	✓	✓	×	×	×
1	3	3	×	×	×	×	×
4	4	4	×	✓	×	×	×
2	5	5	×	×	×	×	×

$$(b) D(\omega^{-1}) = D(\omega)^T$$

$$\Leftrightarrow (\forall (i,j) \in D(\omega^{-1}) \Rightarrow (i,j) \in D(\omega)^T \text{ and } \forall (i,j) \in D(\omega)^T \Rightarrow (i,j) \in D(\omega^{-1})) (*)$$

$$1^\circ \forall (i,j) \in D(\omega^{-1}) \Rightarrow j < \omega^{-1}(i) \text{ and } i < (\omega^{-1})^{-1}(j) = \omega(j).$$

$$\Rightarrow \bar{j} < \omega^{-1}(i) \text{ and } i < \omega(j)$$

$$\Rightarrow (j,i) \in D(\omega).$$

$$\Rightarrow (i,j) \in D(\omega)^T$$

$$2^\circ \forall (i,j) \in D(\omega)^T \Rightarrow (j,i) \in D(\omega)$$

$$\Rightarrow i < \omega(j) \text{ and } j < \omega^{-1}(i).$$

$$\Rightarrow j < \omega^{-1}(i) \text{ and } i < (\omega^{-1})^{-1}(j) = \omega(j).$$

$$\Rightarrow (i,j) \in D(\omega^{-1}).$$

$$\text{by } 1^\circ, 2^\circ (*), D(\omega^{-1}) = D(\omega)^T.$$

(C) Consider a bijection $(i, j) \in D(w) \Leftrightarrow (i, w^{-1}(j)) \in \text{Inv}(w)$

why it is a bijection?

$$\text{Inv}(w) = \{(i, j) : i < j, w(i) > w(j)\}$$

$$1^\circ. (i, w^{-1}(j)) \in \text{Inv}(w) \Rightarrow i < w^{-1}(j) \text{ and } w(i) > w(w^{-1}(j)) = j \\ \Rightarrow (i, j) \in D(w)$$

$$2^\circ (i, j) \in D(w) \Rightarrow j < w(i) \text{ and } i < w^{-1}(j) \\ \Rightarrow w(w^{-1}(j)) < w(i) \text{ and } i < w^{-1}(j) \\ \Rightarrow (i, w^{-1}(j)) \in \text{Inv}(w).$$

by $1^\circ, 2^\circ$ It is a bijection.

$$\text{inv}(w) = \# \text{Inv}(w), \quad \text{inv}(w^{-1}) = \# \text{Inv}(w^{-1})$$

$$\text{by bijection } \# \text{Inv}(w^{-1}) = \# D(w^{-1}).$$

$$\text{by b) } D(w^{-1}) = D(w)^T, \quad \# D(w^{-1}) = \# D(w)^T = \# D(w) \stackrel{\text{by bijection}}{=} \# \text{Inv}(w)$$

$$\text{so } \text{inv}(w) = \text{inv}(w^{-1})$$

P8.

$$(a) w = 123, \quad \text{Des}(w) = \emptyset, \quad \text{maj}(w) = 0, \quad \text{inv}(w) = 0.$$

$$w = 132, \quad \text{Des}(w) = \{2\}, \quad \text{maj}(w) = 2, \quad \text{inv}(w) = 1.$$

$$w = 213, \quad \text{Des}(w) = \{1\}, \quad \text{maj}(w) = 1, \quad \text{inv}(w) = 1$$

$$w = 231, \quad \text{Des}(w) = \{2\}, \quad \text{maj}(w) = 2, \quad \text{inv}(w) = 2$$

$$w = 312, \quad \text{Des}(w) = \{1\}, \quad \text{maj}(w) = 1, \quad \text{inv}(w) = 2$$

$$w = 321, \quad \text{Des}(w) = \{1, 2\}, \quad \text{maj}(w) = 3, \quad \text{inv}(w) = 3.$$

$$\text{when } w = 132, \quad \text{maj}(w) \neq \text{inv}(w).$$

$$(b). w = w_1 w_2 \cdots w_{n-1}.$$

$$\text{Consider } w' = \underline{w_1 w_2 \cdots w_k} \, n \, \underline{w_{k+1} w_{k+2} \cdots w_{n-1}}.$$

$$\text{Let } \alpha = w_1 w_2 \cdots w_k, \quad \beta = w_{k+1} w_{k+2} \cdots w_{n-1}.$$

$$1^\circ w_k < w_{k+1}.$$

$$\text{maj}(w) = \text{maj}(\alpha) + \text{maj}(\beta) + k \# \text{Des}(\beta).$$

$$\text{maj}(w') = \text{maj}(\alpha) + k+1 + \text{maj}(\beta) + (k+1) \# \text{Des}(\beta).$$

$$\text{so } \text{maj}(w') - \text{maj}(w) = k+1 + \# \text{Des}(\beta).$$

$$2^\circ w_k > w_{k+1}.$$

$$\text{maj}(w) = \text{maj}(\alpha) + k + \text{maj}(\beta) + k \# \text{Des}(\beta).$$

$$\text{maj}(w') = \text{maj}(\alpha) + k+1 + \text{maj}(\beta) + (k+1) \# \text{Des}(\beta).$$

$$\text{so } \text{maj}(w') - \text{maj}(w) = 1 + \# \text{Des}(\beta).$$

when $w' = n; w_1 w_2 \dots w_{n-1}$.

$$\text{maj}(w') = 1 + \text{maj}(w) + \# \text{Des}(w).$$

$$\Rightarrow \text{maj}(w') - \text{maj}(w) = 1 + \# \text{Des}(w).$$

when $w' = w_1 w_2 \dots w_{n-1}; n$ $\text{maj}(w') - \text{maj}(w) = 0$.

Define a move satisfies n moves from place k to $k+1$.

$$w_1 w_2 \dots w_k n w_{k+1} \dots w_{n-1} \xrightarrow{\text{move}} w_1 w_2 \dots w_k w_{k+1} n w_{k+2} \dots w_{n-1}.$$

for each move

$$\text{if } w_k < w_{k+1}, \text{ by } 1^\circ, \text{maj}(w') - \text{maj}(w) = k+1 + \# \text{Des}(\beta).$$

$$\text{if } w_k > w_{k+1}, \text{ by } 2^\circ, \text{maj}(w') - \text{maj}(w) = 1 + \# \text{Des}(\beta).$$

Define $\beta_k = w_{k+1} w_{k+2} \dots w_{n-1}$. $w'_k = w_0 w_1 \dots w_k n w_{k+1} \dots w_{n-1}$. $d_k(w) = \text{maj}(w'_k) - \text{maj}(w)$.

$$\text{so } \# \text{Des}(\beta_k) = \begin{cases} \# \text{Des}(\beta_{k-1}), & w_k < w_{k+1} \\ \# \text{Des}(\beta_{k-1}) - 1, & w_k > w_{k+1} \end{cases}$$

$$d_k(w) = \begin{cases} k+1 + \# \text{Des}(\beta_k), & w_k < w_{k+1} \\ 1 + \# \text{Des}(\beta_k), & w_k > w_{k+1} \end{cases}$$

1° now we prove. $i \in \{k: w_k > w_{k+1}\}$, $\{d_i(w)\} = \{1, 2, \dots, \# \text{Des}(w)\}$

every k such that $w_k > w_{k+1}$, $\# \text{Des}(\beta_k)$ will minus 1.

so $d_k(w) = 1 + \# \text{Des}(\beta_k)$ ($w_k > w_{k+1}$) will minus 1

at first $\# \text{Des}(\beta_k) = \# \text{Des}(w)$ at last $\# \text{Des}(\beta_k) = 0$.

so $\{d_i(w)\}_{i \in \{k: w_k > w_{k+1}\}} = \{1, 2, \dots, \# \text{Des}(w)\}$

2° then $i \in \{k: w_k < w_{k+1}\}$, we prove $\{d_i(w)\} = \{\# \text{Des}(w) + 2, \dots, n-1\}$



1° for any k not near the descent.

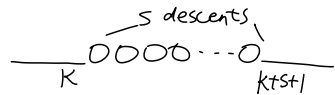
$$d_k(w) = k+1 + \# \text{Des}(\beta_k)$$

when $k \rightarrow k+1$, $\# \text{Des}(\beta_{k+1}) = \# \text{Des}(\beta_k)$. so $d_{k+1}(w) = d_k(w) + 1$.

2° for any k near the descent. suppose there are s continuous descents.

$$d_{k+s+1}(w) = (k+s+1) + 1 + \# \text{Des}(\beta_{k+s+1}).$$

$$\# \text{Des}(\beta_{k+s+1}) = \# \text{Des}(\beta_k) - s$$



$$\Rightarrow d_{k+s+1}(w) = (k+s+1) + 1 + \# \text{Des}(\beta_k) - s = k+2 + \# \text{Des}(\beta_k) = d_k(w) + 1.$$

so each time n moves to next not descent place.

$d_k(w)$ will increase 1.

at first $d_k(w) = 2 + \# \text{Des}(w)$. the first place that n moves to not descent place

at last $d_k(w) = n-1$. the last place that n moves to not descent place.

so $\{d_i(w)\}_{i \in \{k: w_k < w_{k+1}\}} = \{\# \text{Des}(w) + 2, \dots, n-1\}$.

$$\text{so } \{d_i(w)\}_{i \in [n]} = \{d_i(w)\}_{i \in \{k: w_k < w_{k+1}\}} \cup \{d_i(w)\}_{i \in \{k: w_k > w_{k+1}\}} \cup \{d_n(w), d_1(w)\} \\ = \{0, 1, 2, \dots, n-1\}.$$

$$\text{and } \# \{d_i(w)\}_{i \in [n]} = n = \# \{0, 1, 2, \dots, n-1\}$$

so the resulting increases in major index are $0, 1, 2, \dots, n-1$ each occurring once.

$$(C). \text{ Let } f_n(x) = \sum_{w \in S_n} x^{\text{maj}(w)}.$$

by (b). $\forall w \in S_{n-1}$, if we insert n , then $\text{maj}(w)$ will increase $0, 1, 2, \dots, n-1$ in n place in w .
and every $w \in S_{n-1}$ and insert n in every place of w construct S_n .

$$\text{so } f_n(x) = \sum_{w \in S_n} x^{\text{maj}(w)} = \sum_{w \in S_{n-1}} x^{\text{maj}(w)} \cdot (1 + x + x^2 + \dots + x^{n-1}) \\ = f_{n-1}(x) (1 + x + x^2 + \dots + x^{n-1}).$$

$$\text{so } f_n(x) = \frac{f_n(x)}{f_{n-1}(x)} \cdot \frac{f_{n-1}(x)}{f_{n-2}(x)} \dots \frac{f_3(x)}{f_2(x)} \cdot f_1(x). \quad \left. \begin{array}{l} w=12 \quad \text{maj}(w)=0 \\ w=21 \quad \text{maj}(w)=1 \end{array} \right\} \Rightarrow f_2(x) = 1+x \\ = \prod_{k=3}^n (1 + x + x^2 + \dots + x^{k-1}) \cdot (1+x) \\ = \prod_{k=1}^n (1 + x + x^2 + \dots + x^{k-1}).$$

$$\text{so } \sum_{w \in S_n} x^{\text{maj}(w)} = \prod_{k=1}^n (1 + x + x^2 + \dots + x^{k-1}).$$

P9.

$$(a). w = (1, 4, 6) (2, 5) (3).$$

$$b_1=0, b_2=0, b_3=0, \quad 1 \rightarrow 4, b_4=1, \quad 2 \rightarrow 5, b_5=2,$$

$$4 \rightarrow 6, b_6=4.$$

$$\text{encoding} : (0, 0, 0, 1, 2, 4).$$

$$(0, 1, 0, 2, 0, 3)$$

$$1, \quad 1 \rightleftharpoons 2, \quad 1 \rightleftharpoons 2 \quad 3, \quad \begin{array}{c} 1 \rightarrow 2 \quad 3 \\ \swarrow \downarrow \\ \quad 4 \end{array}$$

$$\begin{array}{c} 1 \rightarrow 2 \quad 3 \quad 5 \\ \swarrow \downarrow \\ \quad 4 \end{array}, \quad \begin{array}{c} 1 \rightarrow 2 \quad 3 \quad 5 \\ \swarrow \downarrow \uparrow \downarrow \\ \quad 4 \quad 6 \end{array}$$

$$\text{permutation} : (1, 2, 4) (3, 6) (5)$$

(b). $1^\circ b_k = 0 \Rightarrow k$ is the smallest element in cycle.

if not, $\exists i < k$, there is a cycle containing i and k .

because the number inserts from small to large.

so when k is inserted, there is already an element j which is smaller than k , in that case $b_k = j \neq 0$, contradiction!
and j points to k .

so 1° corrects.

$2^\circ k$ is the smallest element in cycle $\Rightarrow b_k = 0$.

if not, $b_k = j \neq 0$. because the number inserts cycle from small to large, $b_k = j$ means j is inserted before k , so $j < k$

so k is not the smallest element in cycle contradiction!

so 2° corrects.

by $1^\circ 2^\circ$, $b_k = 0$ exactly when k is the smallest element of its cycle in the final.

(c). According to b). $k \in S \Leftrightarrow b_k = 0$.

1° if $1 \notin S$. because 1 is the first element, so $b_1 = 0$.

so there is NO such permutation.

2° if $1 \in S$. then.

if $k \in S$, then $b_k = 0$. 1 choice.

if $k \notin S$, then b_k can be $1, 2, 3, \dots, k-1$, which has $k-1$ choices.

so there are $\prod_{k \notin S} (k-1)$ number of permutations.

In conclusion. if $1 \notin S$, no permutations.

if $1 \in S$, $\prod_{k \notin S} (k-1)$ number of permutations.

P10. a permutation is an involution

(a).^{1°} If there is a cycle have length $n \geq 3$.

$a_1 \xrightarrow{w} a_2 \xrightarrow{w} a_3 \xrightarrow{w} \dots \xrightarrow{w} a_n$, where $a_1, a_2, \dots, a_n$ are all different.
(otherwise, there is a small cycle inside)

so $w(a_1) = a_2$, $w(w(a_1)) = w(a_2) = a_3$.

but $a_1 \neq a_3$, contradiction!

2° If all cycles have length 1 or 2

$\forall i \in [n]$.

i) if i in 1-length cycle, $w(i) = i$

then $w(w(i)) = i$

ii) if i in 2-length cycle $w(i) = j$, $w(j) = i$ $i \neq j$

then $w(w(i)) = w(i) = i$.

so it is an involution.

by 1°, 2°. a permutation is an involution exactly when all its cycles have length 1 or 2.

(b). Consider n in which cycle.

by (a), n should be in cycle with length 1 or 2.

1° n in length 1 cycle, then for the rest $n-1$ numbers,

there are I_{n-1} ways of involutions.

2° n in length 2 cycle, suppose it is (n, i) , ($i = 1, 2, \dots, n-1$)

and the rest of $n-2$ numbers has I_{n-2} ways of involutions. $\rightarrow$ it has $n-1$ ways to choose i

So there are $(n-1) \cdot I_{n-2}$ ways.

Combine 1°, 2°, for S_n , there are $I_{n-1} + (n-1) I_{n-2}$ ways of involutions.

so $I_n = I_{n-1} + (n-1) I_{n-2}$. When $n=1$, $w = (1)$ is an involution. so $I_1 = 1$

(c). A permutation w is involution $\Leftrightarrow$ All cycles have length 1 or 2 (by (a)).
 $\Leftrightarrow w = w^{-1} (*)$

1° if all cycles have length 1 or 2.

i) $w(i) = i \Rightarrow w^{-1}(i) = i = w(i)$ ii) $w(i) = j$, $w(j) = i \Rightarrow w(i) = j = w^{-1}(i)$.
 $w(j) = i = w^{-1}(j)$

2° if $w = w^{-1}$

if there exist a cycle have length $n \geq 3$.

$a_1 \rightarrow a_2 \rightarrow \dots \rightarrow a_n$ $a_1, a_2, \dots, a_n$ are all different.

$w(a_1) = a_2, w^{-1}(a_1) = a_n, a_2 \neq a_n \Rightarrow w \neq w^{-1}$ contradiction!
so (*) corrects.

We find permutation w but is not involution.

by (*) $w \neq w^{-1}$, so we can pair $\{w, w^{-1}\}$. $w \neq Id$

because $(w^{-1})^{-1} = w$, so we can pair all that permutation.

$\{w_1, w_1^{-1}\}, \{w_2, w_2^{-1}\}, \dots, \{w_k, w_k^{-1}\}$, and all of them are different.

so $n! - I_n$ is even, when $n \geq 2$, $n!$ is even

so I_n is even, remove the trivial involution $w = (1)(2) \dots (n)$.

then the nontrivial involution is S_n is odd when $n \geq 2$.